Table 1: Implementation details including training stages, datasets, and essential parameters: "BS" for batch size, "Length" for total sequence length. "Graphist" is trained on academic data; "Graphist*" on proprietary data. "PE." denotes the Patch Embedding Layer. "Cap." signifies image captioning task. "Vari." denotes the task variants in Section 5.1. "IH-RGBA" is in-house image-text dataset, where images with alpha channel. "IH-Design" is 80k in-house Graphic Design Dataset like Crello [86].

Stage	BS	Length	Tune Part	Task	Training datasets	
					Graphist	Graphist*
1	128	1536	PE Projector	Cap. HLG	ShareGPT4v [14] Flickr30k [66] Crello [86]	+ IH-RGBA
2	64	2048	Projector LLM	Cap. HLG	ShareGPT4v [14] Flickr30k [66] Crello [86]	+ IH-RGBA + IH-Design
3	64	3584	Projector LLM	Cap. HLG GLG Vari.	ShareGPT4v [14] Flickr30k [66] Crello [86] CGL-V2 [46]	+ IH-RGBA + IH-Design

MLP layer to map the $\mathbf{V}$ to $\mathbf{V}' \in \mathbb{R}^{5\times D'}$ for modal alignment and dimension matching of LLM, where D' is the word embedding dimension defined by LLM. Visual embedding can be inserted into anywhere of input sequence. Inspired by Shikra [13], we employ the digital tokens to represent coordinates, devoid of any vocab, specialty encoders, or any pre-/post-detectors for encoding position information. Model variants include Graphist-Tiny utilizing Qwen1.5-0.5B and Graphist-Base engaging Qwen1.5-7B.

4.2 Training Strategy

The proposed Graphist is trained in three stages. Initially, Stage-1 targets the patch embedding layer and the projector layer. The focal training tasks during this phase predominantly encompass image captioning and the HLG task, leveraging a substantial batch size coupled with a reduced sequence length for efficiency. The primary objectives is to calibrate the visual encoder to interpret the alpha channel in RGB-A imagery and to align the projector with both visual and linguistic features.

Progressing to Stage-2, the training expands to encompass both the projector layer and the full LLM. Training tasks from the initial stage are retained; however, the HLG task is administered with greater frequency to highlight the model's fundamental understanding of graphic layout.

Finally, in Stage-3, maintains the focus on the projector layer and the complete LLM, but aiming to adapt the model to a broader range of graphic design task types. Traditional GLG task is also regarded as a specific task type in this stage. We list primary configurations during training in Table 1.

To train the model to arrange inputs layer order and spatial coordinates, we randomly shuffle the input elements with a 0.75 probability for all of the Graphic Design datasets used in the training process. In these three stages, the first stage trains 10k steps, while the second and third stages both train 20k steps. The training for the Graphist-Base is over the course of 5 days, whereas for the Graphist-Tiny model, a shorter duration of 2 days is anticipated.

5 Experiment

5.1 Datasets

Crello. [86] Crello dataset [1] furnishes an array of graphic compositions derived from a web-based design utility, namely Crello (Now change the name to VistaCreate[2]). It covers an extensive range of graphic compositions suited for various applications such as social media infographics, digital banner ads, blog headers, and printed poster templates. Inside the dataset, each graphic composition includes detailed information about the layering order, spatial positioning, and categorical details of the design elements. The dataset is advantageous for tasks like GLG and has been the foundation for several methods [36, 86]. Additionally, it is a good playground for HLG task. In Flex-DM [36], the dataset is partitioned into 19,095 training, 1,951 validation, and 2,375 testing examples. However, they used the Crello v2, but since the current version released by Crello is v4, we used the intersection of all parts in the two version test sets, a total of 242 graphic compositions as the test set in experiments.

CGL-Dataset V2 is an extension of the CGL-Dataset [102] and provides a dataset for generating advertising poster layouts, including 60,548 training samples and 1,035 testing samples. Previous work [46] has built two tasks based on this dataset: the Content-Aware task and the Content-Agnostic task. In the Content-Aware task, the model is challenged with an input image and textual content, and is required to output the positions of all textual content in the image while autonomously adding underlay and decorative information. The Content-Agnostic task, on the other hand, only requires an image as input. In this case, the model autonomously determines the output, which includes text, underlay, decoration, and other relevant layering information. Graphist has integrated these tasks alongside HLG/GLG tasks to facilitate comparison across a broader spectrum of layout methods [35, 43, 46, 47, 101].

5.2 Evaluation Metrics

Inverse order pair ratio (IOPR). The appropriate order of layers is crucial for the results of HLG. We develop IOPR, which is a ratio representing the

[1] https://huggingface.co/datasets/cyberagent/crello
[2] https://create.vista.com/

fraction of overlap element pairs that are in inverse order according to the model's predictions out of all possible overlapping element pairs. It is calculated as

$$\text{IOPR} = \frac{\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \mathbb{1}\left(\mathcal{O}_j < \mathcal{O}_i \wedge \mathbb{1}\left(i,j\right)\right)}{\sum_{i=0}^{n-1} \sum_{j=i+1}^{n} \mathbb{1}}, \tag{1}$$

where n is the number of layers in the hierarchical structure. $\mathbb{1}$ is an indicator function that returns 1 if the argument condition is true and 0 otherwise. $\mathcal{O}$ denotes the output order or predicted order of the layers as determined by the model. $\mathcal{O}_i$ and $\mathcal{O}_j$ correspond to the predicted order positions of the ith and j^{th} layers, respectively. $\texttt{overlap}(i,j)$ is a predicate function that determines whether the ith and j^{th} layers overlap. A low IOPR would suggest that the model is quite accurate in predicting the correct order of layers because there are few inverse order pairs. Contrastingly, a high IOPR would suggest the model often predicts the wrong sequence, indicating lower prediction accuracy. In our experiments, we employed IOPR_{min} and IOPR_{avg} to evaluate the performance of the model, representing respectively the average score and the minimum score in the test dataset. While the average IOPR across a dataset can provide significant insights, it should be noted that variances in layer orders for specific graphic compositions, when compared to ground truth, do not inherently imply subpar layout performance.

GPT-4V Eval. In addition to the layers order, the overall aesthetic quality and harmony of the elements in the graphic composition are vitally important. We utilize GPT-4V to evaluate our approach and compare it with other alternatives. Following COLE [38], we use four scores named GPT-4V rating including S_{DL}, S_{GI}, S_{IO} and S_{TV} use GPT-4V to evaluate the quality of the graphic composition we generate.

- S_{DL} means the graphic design should present a clean, balanced, and consistent layout. The organization of elements should enhance the message, with clear paths for the eye to follow.
- S_{GI} reflects that any graphics or images used should enhance the design rather than distract from it. They should be high quality, relevant, and harmonious with other elements.
- S_{IO} evaluates the innovation level of the design.
- S_{TV} represents text readability. A lower score would be assigned if the readability of the text is poor due to the color of the text being similar to the background color or overlapping of the text.

As a supplement to GPT-4V rating [38], we propose GPT-4V voting. Here, GPT-4V also partakes in a comparative analysis. It selects the most proficient graphic composition when confronted with two competing outputs. This preference distribution acts as a testament to the discernibility of GPT-4V in recognizing and preferring one method over the other across a range of comparative samples. GPT-4V rating [38] and GPT-4V voting together form the GPT-4V Eval for evaluating layout ability.